

Harvard University
Physics 141 Pset 5

Due October 24, 2025, 5 PM

1. In this problem, we will re-derive results regarding Brownian motion that you studied in lecture. Consider a particle of mass m in a viscous fluid. The particle's position in 3D is represented by $\mathbf{x}(t)$, and its velocity is denoted by $\mathbf{v}(t)$. Suppose the particle has initial position $\mathbf{x}_0$ and initial velocity $\mathbf{v}_0$. For now, assume the only force acting on the particle is drag. The equation of motion is given by

$$m \frac{d\mathbf{v}}{dt} = -\zeta \mathbf{v}(t)$$
$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}(t)$$

Where ζ is some constant. For a spherical particle, one can write $\zeta = 6\pi\mu R$, where μ denotes the viscosity of the fluid, R the radius of the particle (this is called Stokes drag).

- (a) Show that, for the above equations of motion, $\mathbf{v}(t) = \mathbf{v}_0 e^{-\frac{t}{\tau}}$, where $\tau = \frac{m}{\zeta}$.
- (b) Provide a rough estimate for the velocity relaxation time τ for a typical *E. coli* particle (assume spherical particle in water). Show via dimensional analysis that your answer has the correct units.

For now, our equations of motion only factor in the effect of drag on the particle. However, as the particle sits in the fluid, it is being constantly bombarded by solvent molecules in the fluid. These collisions occur at a longer timescale compared to τ and can be described via a “white noise” term $\eta(t)$. White noise has the property that $\langle \eta(t) \rangle = 0$ and that for two given times t and t'

$$\langle \eta(t) \cdot \eta(t') \rangle = \Gamma \delta(t - t')$$

Recall that the Kronecker delta function is defined as $\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$.

The fact that the mean is zero tells you that the collisions have no preferential direction, and the second condition on $\eta(t)$ tells you that white noise has no memory of its past. The constant Γ is called the fluctuation intensity. We can naively add this white noise term to our equations of motion, resulting in

$$m \frac{d\mathbf{v}}{dt} = -\zeta \mathbf{v}(t) + \eta(t)$$

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}(t)$$

These are not deterministic equations anymore. We call these stochastic differential equations (specifically the Langevin equations, as you saw in class) due to their dependence on the noise term $\eta(t)$. We can naively write solutions to these equations as if they were typical ordinary differential equations. We can then extract the physics by taking averages over multiple instances of noise $\eta(t)$ and studying the statistics of these solutions. For example, a solution to the first equation is

$$\mathbf{v}(t) = \mathbf{v}_0 e^{-\frac{t}{\tau}} + \frac{1}{m} \int_0^t ds e^{-\frac{t-s}{\tau}} \eta(s)$$

- (c) Verify that the above equation is a solution to the equation of motion for $\mathbf{v}(t)$.
- (d) Show that $\langle \mathbf{v}(t) \rangle = \mathbf{v}_0 e^{-\frac{t}{\tau}}$ (the solution to the deterministic equation you showed in (a)).
- (e) Show that the mean squared velocity is given by

$$\langle \mathbf{v}(t) \cdot \mathbf{v}(t) \rangle = v_0^2 e^{-\frac{2t}{\tau}} + \frac{\Gamma}{2m\zeta} (1 - e^{-\frac{2t}{\tau}})$$

- (f) Suppose $t \gg \tau$, what is the mean squared velocity in this limit? Explain why there is no dependence on $\mathbf{v}_0$.

For the remainder of this problem, we will now assume that $x(t)$ and $v(t)$ are variables in 1D (one translational degree of freedom). We will also solve the remainder of the problem in the overdamped limit $t \gg \tau$.

- (g) Use the equipartition theorem to show that for $t \gg \tau$, we have $\Gamma = 2\zeta k_b T$.

The equation you derived in (g) is a representation of the fluctuation-dissipation theorem. We can rewrite the answer in (g) as

$$\langle \eta(t) \cdot \eta(t') \rangle = 2\zeta k_b T \delta(t - t')$$

The term on the left side represents the fluctuation (random noise) and the term on the right side represents the dissipation. One way to interpret this formula is that for a system with a noisy process, there is a dissipation term. The noise and dissipation

both cause one another and are both caused by the same underlying physics. For example, in the context of Brownian motion, the fluctuation (the random movement of the particle) and the dissipation (drag force on particle) are caused by the same underlying force (collisions with solvent molecules in the fluid).

Moreover, the random movement of these solvent molecules can be associated with the effect of dissipation. When your large particle gets perturbed, drag force acts on it, resulting in its kinetic energy being turned into heat. This heat contributes to thermal fluctuations. These thermal fluctuations explain the random motion of the solvent molecules, which in turn perturb the larger particle.

- (h) We will now find $x(t)$ in the overdamped limit. Assume $t \gg \tau$, and assume that $x_0 = 0$. Show that

$$x(t) = \int_0^t ds \frac{\eta(s)}{\zeta}$$

Hint: First, take your formula for $v(t)$ and write $m = \zeta\tau$. Then, use the fact that $\lim_{\tau \rightarrow 0} e^{\frac{-t}{\tau}} = \delta(t)$. Finally, use $\frac{dx}{dt} = v(t)$ and integrate.

- (i) Show that $\langle x(t) \rangle = 0$
(j) Show that

$$\langle x^2(t) \rangle = 2 \frac{k_b T}{\zeta} t$$

As you saw in class, we define the diffusion coefficient $D = \frac{k_b T}{\zeta}$, in which case $\langle x^2(t) \rangle = 2Dt$

The formula for D is another way to think of the fluctuation-dissipation theorem. The macroscale quantity D can be represented as a ratio of two microscopic quantities. The numerator is $k_b T$, a representation of the thermal fluctuations at microscopic scale. The denominator is ζ , a representation of the effect of drag (the microscopic collisions with the solvent molecules that slow it down).

We will now explore a similar system that you studied in class. Consider the orientation of a spherical particle of radius R in a viscous fluid of (dynamic) viscosity μ . Let $\omega(t)$ denote the angular velocity about a fixed axis. The rotational drag for a sphere is $\zeta_r = 8\pi\mu R^3$. Let I be the moment of inertia (for a solid sphere $I = \frac{2}{5}mR^2$).

The rotational Langevin equations are

$$I \frac{d\omega}{dt} = -\zeta_r \omega(t) + \xi(t), \quad \frac{d\theta}{dt} = \omega(t)$$

with a zero-mean white-noise torque $\xi(t)$ satisfying

$$\langle \xi(t) \rangle = 0, \quad \langle \xi(t) \xi(t') \rangle = \Gamma_r \delta(t - t')$$

- (k) Compare the Langevin equations for translation and rotation. Explain in words the equivalence between each term (for example, explain why mass m is replaced with a moment of inertia I).
- (l) You will notice that the Langevin equations for translation and rotation are similar. Using your result from (j), show that $\langle \theta^2(t) \rangle = 2D_r t$, where

$$D_r = \frac{k_b T}{8\pi\mu R^3}$$